EC441 Midterm 2 Study Guide (Condensed)

Network Layer (L13)
Addressing, forwarding, routing. Forwarding = per-packet decision using table. Routing = global computation of paths. Best-effort, connectionless design.

IP Addressing + CIDR (L14)
IP = 32 bits. Split into network + host. CIDR /N defines prefix. Hosts = $2^{(32-N)}-2$. Network = IP AND mask. Enables aggregation.

Subnetting
Divide networks by extending prefix. Example: /16 → /18 gives 4 subnets. Each subnet size = $2^{(32-\text{prefix})}$.

Routing Algorithms (L15)
Dijkstra (link-state): full topology, compute shortest path tree. Cost = sum of weights. Used in OSPF.

Distance Vector (L16)
Bellman-Ford: $d_x(y)=\min(c(x,v)+d_v(y))$. Distributed, iterative. Problems: count-to-infinity, slow convergence.

IPv4 + NAT (L17)
TTL prevents loops. NAT maps private → public IP using ports. ICMP used for errors (ping, traceroute).

Transport Layer + RDT (L18)
UDP: no reliability. Stop-and-Wait inefficient due to idle time. GBN: retransmit all after loss. SR: selective retransmission.

TCP Basics (L19)
3-way handshake: SYN → SYN-ACK → ACK. Reliable byte stream using sequence numbers + ACKs. Flow control via rwnd.

TCP Congestion Control (L20)
cwnd controls sending rate. Slow start (exponential). AIMD: +1 MSS/RTT, halve on loss. Timeout → reset to 1.

Key Formulas
Hosts = $2^{(32-N)}-2$ | Throughput $\approx 1/(RTT\sqrt{p})$ | Bellman-Ford equation